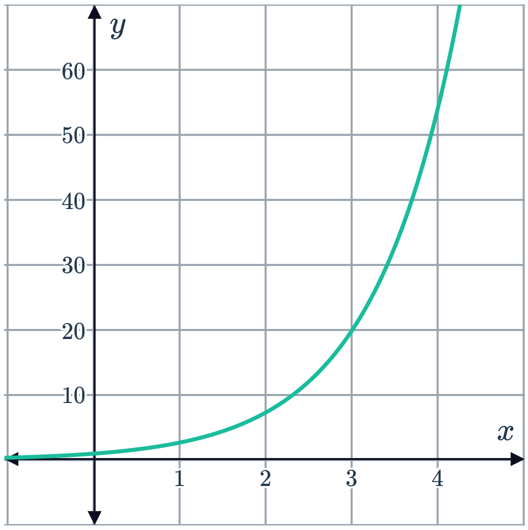
Approximate areas under graphs

1 The following graph shows the function $y = e^x$:



a Find an estimate for the area between the function and the x-axis for $0 \le x \le 4$, giving your answer to three decimal places and using:

i The left-endpoint approximation with 4 rectangles.

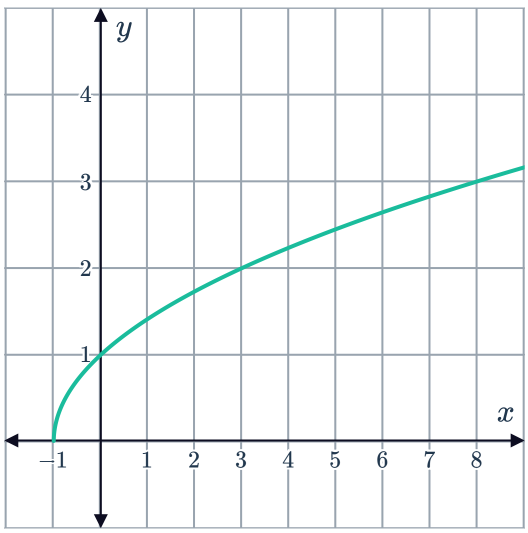
ii The right-endpoint approximation with 4 rectangles.

iii Technology and the left-endpoint approximation with 50 rectangles.

b Use integration to find the exact area between the function and the x-axis for $0 \le x \le 4$.

c Calculate the percentage error of the approximation in (a) part (iii) compared to the exact area. Round your answer to the nearest percent.

2 The following graph shows the function $y = \sqrt{x+1}$:



a Find an estimate for the area between the function and the x-axis for $2 \le x \le 6$, giving your answer to three decimal places and using:

i The left-endpoint approximation with 4 rectangles.

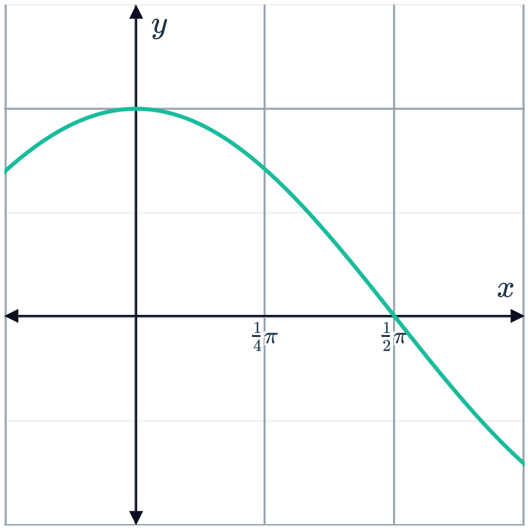
ii The right-endpoint approximation with 4 rectangles.

iii Technology and the right-endpoint approximation with 40 rectangles.

b Use integration to find the area between the function and the x-axis for $2 \le x \le 6$. Round your answer to four decimal places.

c Calculate the percentage error of the approximation in (a) part (iii) compared to area calculated by integration in part (b). Round your answer to two decimal places.

3 The following graph shows the function f [illegible] $\cos x$:



a Find an estimate for the area between the function and the x-axis for $0 \leq x \leq \frac{\pi}{2}$, giving your answer to three decimal places and using left-endpoint approximation with:

i 3 rectangles.

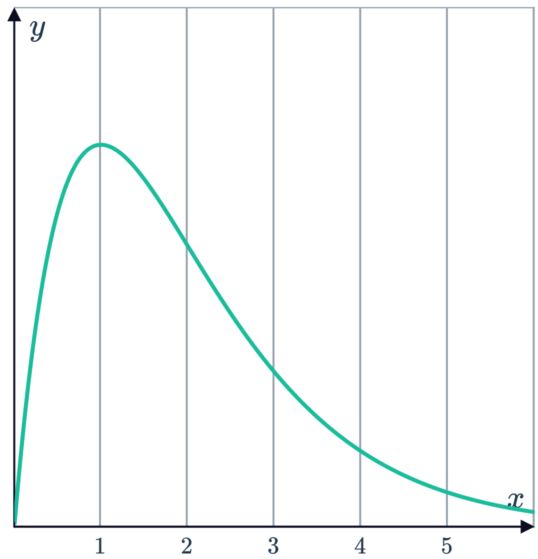
ii 5 rectangles.

iii Technology and 50 rectangles.

b Use integration to find the exact area between the function and the x-axis for $0 \leq x \leq \frac{\pi}{2}$.

c Calculate the percentage error of the approximation in (a) part (iii) compared compared to the exact area. Round your answer to the nearest percent.

4 Consider the function $f(x) = xe^{-x}$ and the given graph of $y = f(x)$:



a Find $f'(x)$.

b Hence, show that:
$\int xe^{-x} = -xe^{-x} - e^{-x} + c$.

c Calculate an overestimate for the area bounded by $f(x)$ and the x-axis for $1 \leq x \leq 5$, giving your answer to three decimal places and using:

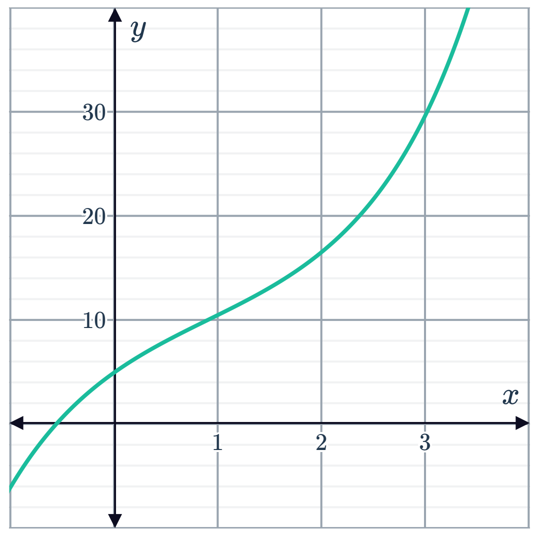
i 2 rectangles of equal width.

ii 4 rectangles of equal width.

d Find the exact area bounded by $f(x)$ and the x-axis over $1 \leq x \leq 5$, using part (b).

e Calculate the percentage error for the two estimates given in part (c). Round your answer to two decimal places.

5 Consider the function $f(x) = e^x + 10 - 6e$ [redacted] d the given graph of $y = f(x)$:

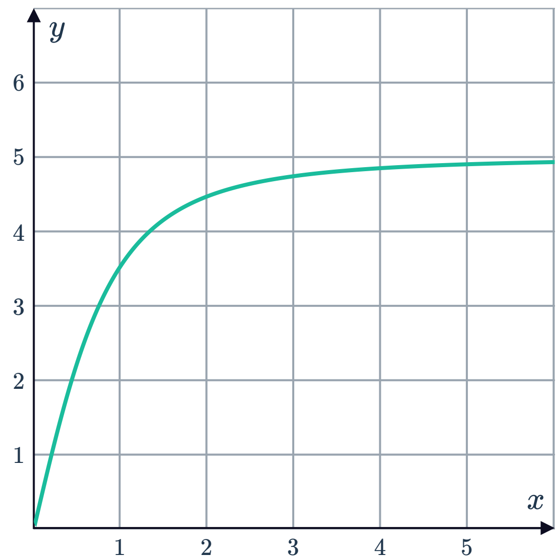


a Calculate an overestimate for the area bounded by $f(x)$ and the x-axis for $1 \leq x \leq 3$ using 4 rectangles of equal width. Give your answer to four decimal places.

b Calculate an underestimate for the area bounded by $f(x)$ and the x-axis for $1 \leq x \leq 3$ using 4 rectangles of equal width. Give your answer to four decimal places.

c Given $f''(x) > 0$ for $x > 0.9$, which estimate will be closer to the exact area bounded by $f(x)$ and the x-axis for $1 \leq x \leq 3$. Explain your answer.

6 Consider the function $f(x) = \dfrac{5x}{\sqrt{x^2+1}}$ and the given graph of $y = f(x)$:



a Calculate an overestimate for the area bounded by $f(x)$ and the x-axis for $0 \leq x \leq 2$ using 2 rectangles of equal width. Give your answer to three decimal places.

b Calculate an overestimate for the area bounded by $f(x)$ and the x-axis for $2 \leq x \leq 4$ using 2 rectangles of equal width. Give your answer to three decimal places.

c Which of the estimates calculated in (a) and (b) will be closer to the exact area being estimated? Explain your answer.

d Would an underestimate or overestimate using 4 rectangles of equal width provide a closer estimate to the exact area bounded by $f(x)$ and the x-axis for $0 \leq x \leq 4$. Explain your answer.

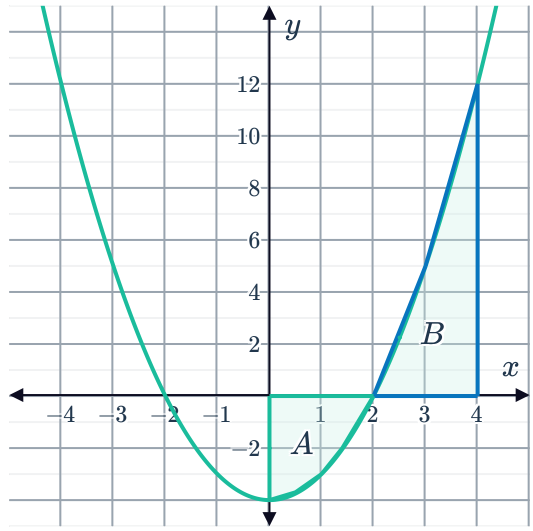
Definite integration

7 The following graph shows the function $y = x^2 - 4$:

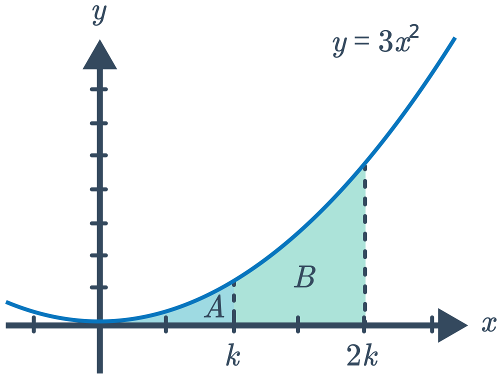
a Find the area A.

b Find the area B.

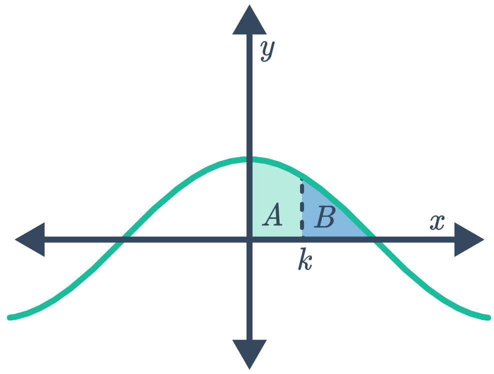
c Find the ratio of the areas $A : B$.



8 The following graph shows the function $y = 3x^2$:
Let A define the area under the curve over the interval $[0, k]$ and B define the area under the curve over the interval $[k, 2k]$.
Find the ratio of the areas $A : B$.

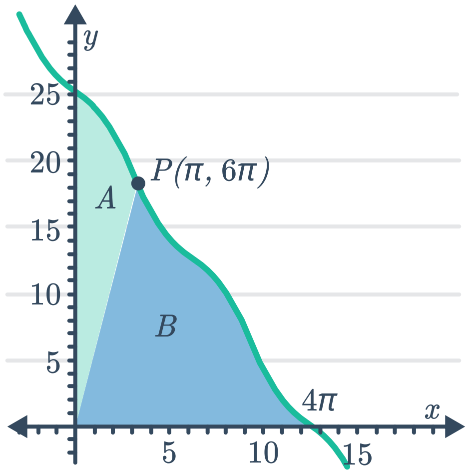


9 The following graph shows the function $y = \cos x$:
Find k such that ratio $A : B$ is $1 : 1$.



10 Consider the function $f(x) = e^x$. Let A define the area under the curve over the interval $[0, k]$ and B define the area under the curve over the interval $[0, 2k]$. Find the relationship between A and B.

11 The curve $y = 8\pi - 2x + \sin x$ is shown passing through the point $P(\pi, 6\pi)$. A straight line joins the origin to point P.



a Find the value of $\int_0^{\pi} (8\pi - 2x + \sin x)\, dx$.

b Find the value of A.

c Find the value of B.

d Determine the ratio $A : B$ in the form $1 : k$. Round k to two decimal places.

12 Consider the function $f(x) = 6ax - 6x^2$, for $a \geq 0$.

a For $a = 1$, sketch the function.

b Find the area bound by the curve and the x-axis.

c Use your calculator to complete the table of the area bound by the graph and the x-axis for different values of a.

a	1	2	3	5	10
Area $(\text{units})^2$					

d Make a conjecture for the value of the area bounded by $f(x) = 6ax - 6x^2$ and the x-axis, for $a \geq 0$.

e Prove your conjecture.

13 Consider the function $f(x) = kx^n(1-x)$ [illegible]e interval $[0,1]$, where n is a positive integer.

a For $n = 1$, find the value of k if the area under the curve on the interval $[0,1]$ is 1 unit^2.

b Using technology, find the area under the curve $y = x^2(1-x)$ on the interval $[0,1]$.

c Hence, find the value of k that would create an area of 1 unit^2 under the curve of $f(x)$ over the interval $[0,1]$, when $n = 2$.

d Complete the following table showing the value of k required for different values of n, if the area on the interval $[0,1]$ is 1 unit^2:

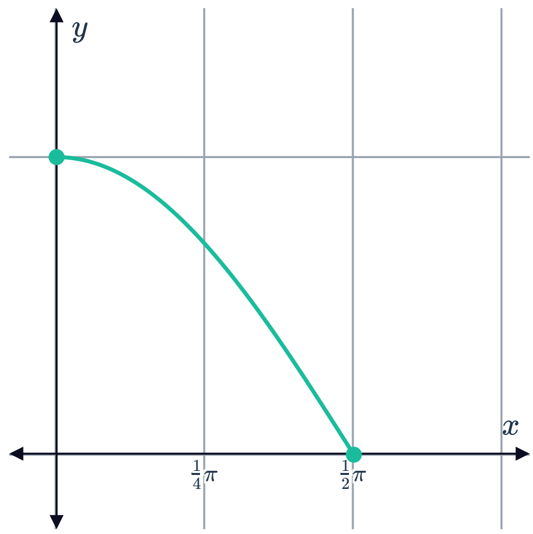
n	1	2	3	4	5
k					

e Make a conjecture for the value of k in terms of n, required if the area under $f(x)$ on the interval $[0,1]$ is 1 unit^2.

f Prove your conjecture.

14 Consider the curve $y = f_k(x)$ where $f_k(x) = \cos(kx)$ and k is a positive integer. The domain of the function $f_k(x)$ is dependent on k which is $0 \le x \le \dfrac{\pi}{2k}$.

a The given graph is of $f_1(x)$, where $k = 1$, thus $f_1(x) = \cos(1x)$ and the domain is $0 \le x \le \dfrac{\pi}{2}$.
Find the area between $y = f_1(x)$ and the x-axis over the domain of the function.



b Consider $y = f_k(x)$ for $k = 2$:

i State the function for $f_2(x)$.

ii State the domain of $f_2(x)$.

iii Find the area between $y = f_2(x)$ and the x-axis over the domain of the function.

c Find the area under $f_3(x)$ over its domain.

d Make a conjecture for the value of the area bounded by $y = f_k(x)$ and the x-axis over $0 \le x \le \dfrac{\pi}{2k}$, for any positive integer value of k.

e Prove your conjecture.